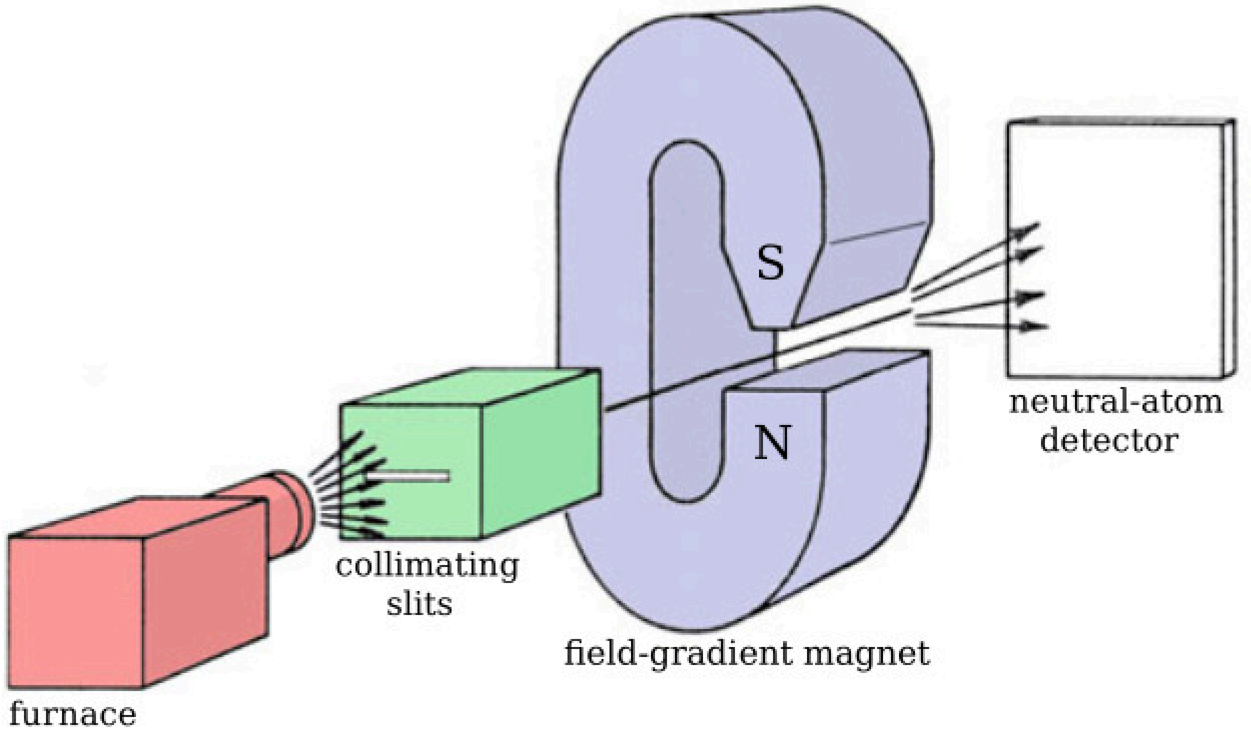
Stern-Gerlach Experiment

Luca Cicu - Lorenzo Liuzzo

27/11/2025

Experiment description

Apparatus


S
N
neutral-atom detector
collimating slits
field-gradient magnet
furnace

Angular & Magnetic momenta

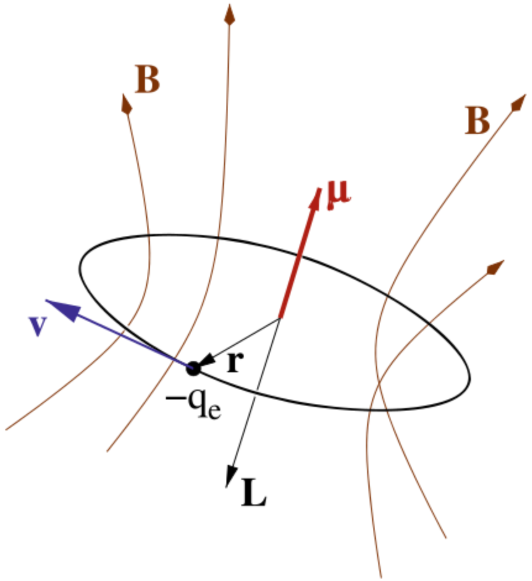
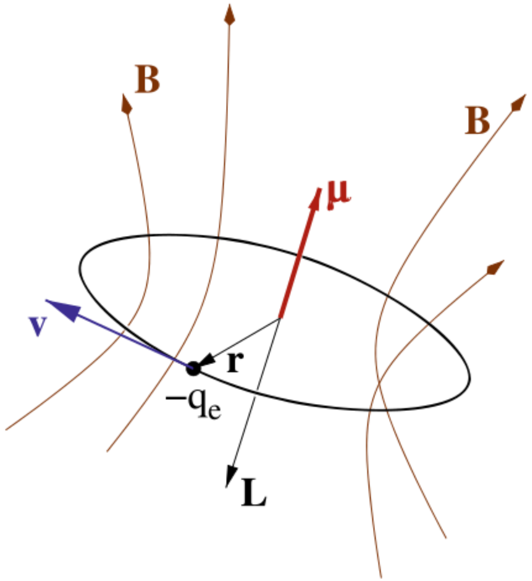


Figure 2: **Angular** and **magnetic** momenta generated by an electron orbiting circularly.

Angular & Magnetic momenta



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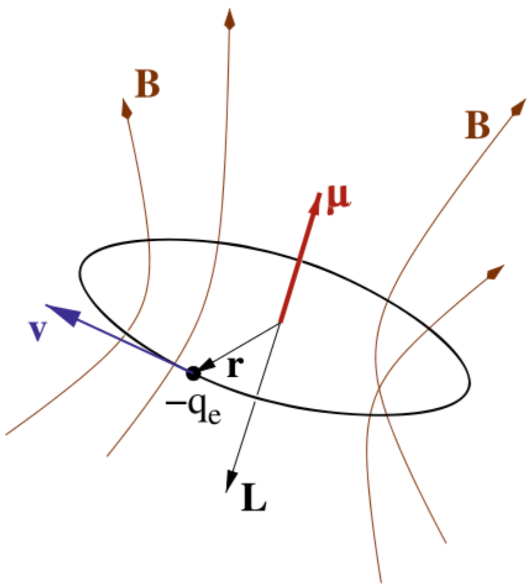
Angular momentum $\boldsymbol{L} = \boldsymbol{x} \times \boldsymbol{p}$

Magnetic momentum $\boldsymbol{\mu} = I \Sigma \hat{\boldsymbol{n}}$

Angular & Magnetic momenta



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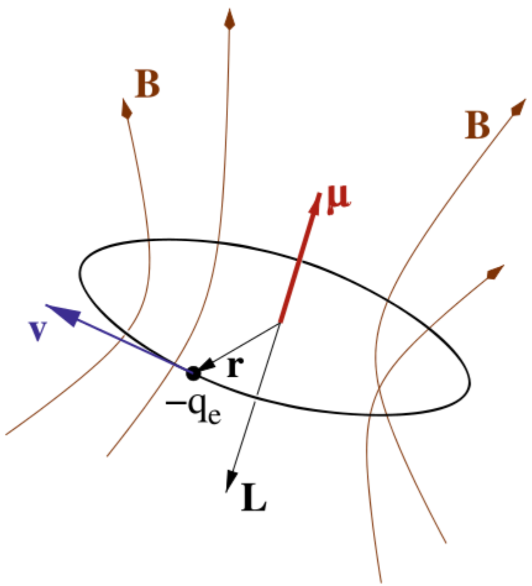

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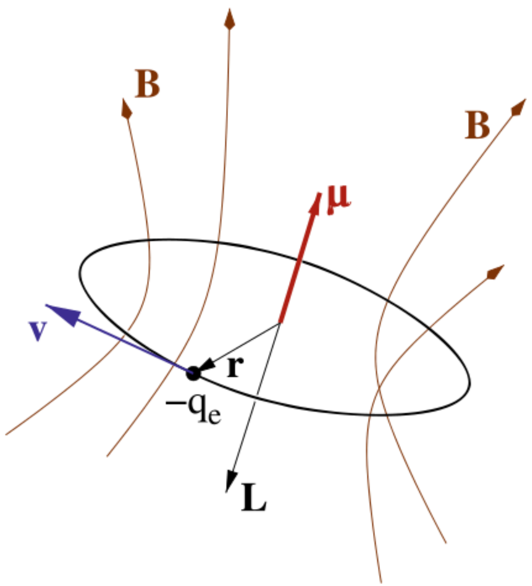

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The **interaction** between $\boldsymbol{\mu}$ and an **external magnetic field** B is represented by the magnetic hamitonian

$$H_{\text{magn}} = -\boldsymbol{\mu} \cdot \boldsymbol{B}$$

Magnetic Interaction

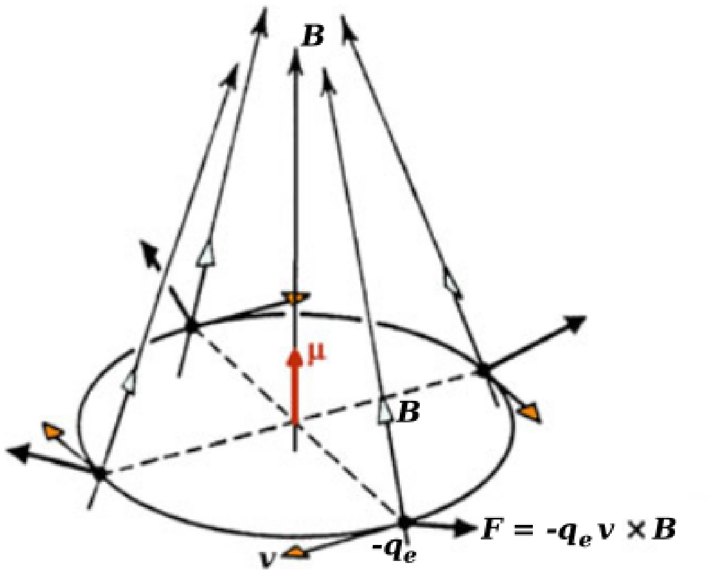

Figure 3: Microscopic origin of the magnetic force.

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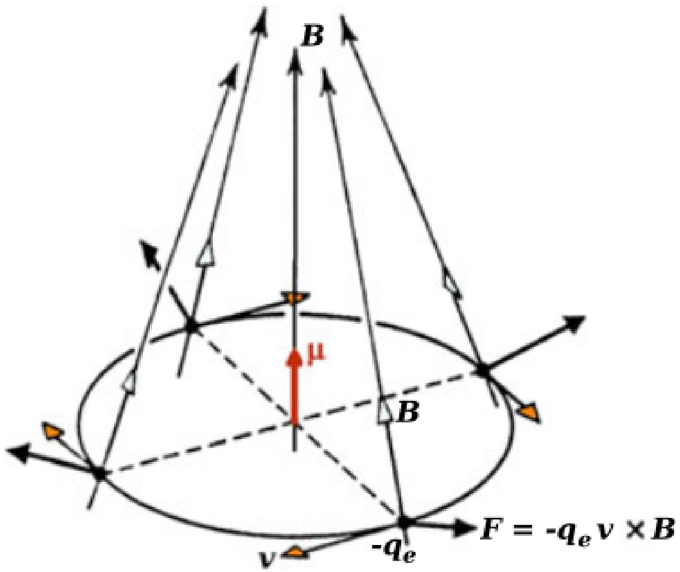

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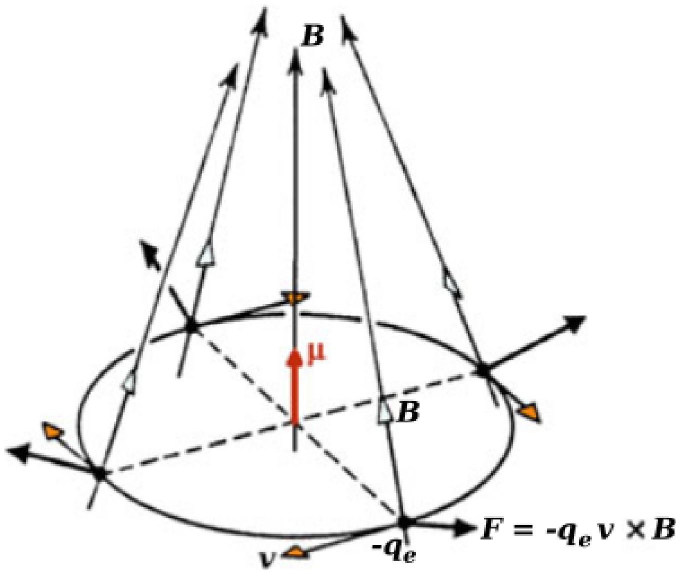

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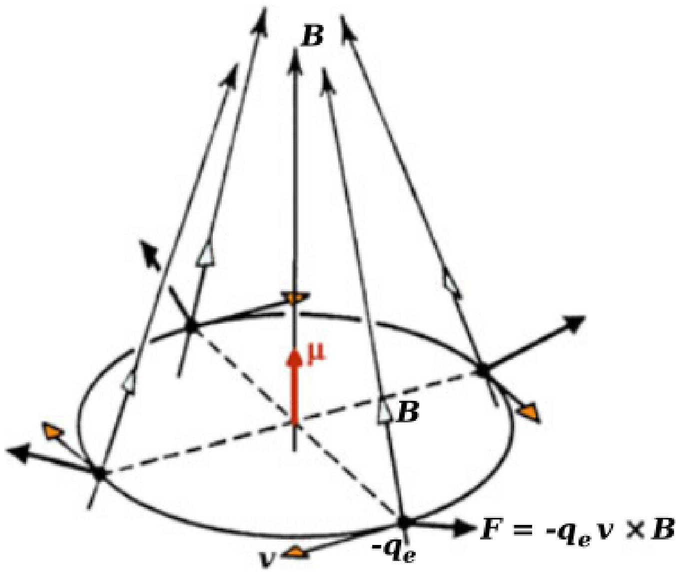

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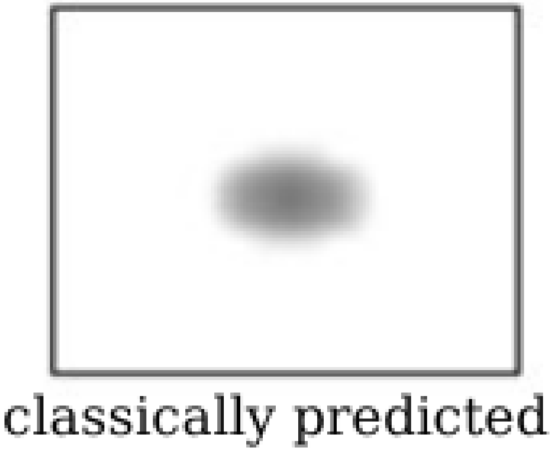
$$z(t) = z_0 + \dot{z}_0 t + \frac{\mu_z}{2m} \frac{\partial B_z}{\partial z} t^2$$

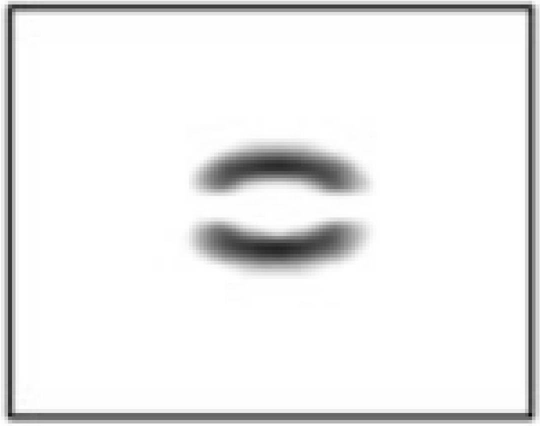
Experimental Results

Classical Results



Diocane

Observed Results



observed

Why two deflected spots?

Historical Context

Historical Context

Planck quanti

Hello world

einstein photoe

Hello world

bohr atom

Hello world

Quantum Mechanics

QM — States

A quantum state is a vector $|\psi\rangle$ in a **Hilbert** space, that is a complete inner product space with normalized elements.

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$$\langle\varphi|\psi\rangle = \sum_i \langle\varphi|e_i\rangle\langle e_i|\psi\rangle = \sum_i \left(c_i^{\varphi}\right)^* c_i^{\psi}$$

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- **Born's rule**

$$\langle\psi|\psi\rangle = \sum_i \left(c_i^\psi\right)^* c_i^\psi = \sum_i |c_i^\psi|^2 = 1$$

$|c_i^\psi|^2$ is the **probability** to find $|\psi\rangle$ in $|e_i\rangle$

QM — Operators

A **linear operator** $\hat{O}$ is a transformation from an Hilbert space into itself, that acts like

$$\hat{O}(\alpha|\psi\rangle + \beta|\varphi\rangle) = \alpha\hat{O}|\psi\rangle + \beta\hat{O}|\varphi\rangle = \alpha|\psi'\rangle + \beta|\varphi'\rangle$$

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The action of $\hat{O}$ on a state can be entirely represented by its action on some **basis**:

$$|\psi'\rangle = \hat{O}|\psi\rangle = \sum_i \hat{O}|e_i\rangle\langle e_i|\psi\rangle = \sum_{ij}|e_j\rangle\langle e_j|\hat{O}|e_i\rangle\langle e_i|\psi\rangle = \sum_{ij}|e_j\rangle O_{ij}c_i^{\psi}$$

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We can also define the **adjoint** operator $\hat{O}^\dagger$: $\quad \langle\psi'| = \langle\psi|\hat{O}^\dagger \quad \hat{O}^\dagger_{ij} = \left(\hat{O}_{ji}\right)^*$

QM — Operators & Observables

We can associate every **observable** of a system to an operator

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where λ_i are the **possible results** and $|e_i\rangle$ are the **possible states** in which the system is founded **after** the measurement.

The requirements are that, **indipendently by the choice of the basis**,

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Such operators are called **hermitian** and satisfy $\quad \hat{O}^\dagger_{ij} = \lambda_i^* \delta_{ij} = \lambda_i \delta_{ij} = \hat{O}_{ij}$

QM — Commutator of Operators

We define the **commutator** between two operators $\hat{A}$ and $\hat{B}$ as

$$\left[\hat{A}, \hat{B}\right] = \hat{A}\hat{B} - \hat{B}\hat{A} = -\left[\hat{B}, \hat{A}\right]$$

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Two observables can be **simultaneously** determined iif their operators are compatible, that is if they share a **common basis** of eigenstates:

$$\hat{A}|e_i\rangle = \lambda_i|e_i\rangle \quad \hat{B}|e_i\rangle = \mu_i|e_i\rangle \quad \Leftrightarrow \quad \langle e_i|\left[\hat{A}, \hat{B}\right]|e_j\rangle = \left(\lambda_i\mu_i - \lambda_j\mu_j\right)\langle e_i|e_j\rangle = 0$$

Position and impulse operators don’t commute!

$$[\hat{p}, \hat{x}] = -i\hbar$$

QM — Angular Momentum

Definition

$$\hat{L}_k = (\hat{\boldsymbol{x}} \times \hat{\boldsymbol{p}})_k = \varepsilon_{kij}\hat{x}_i\hat{p}_j$$

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$$\hat{\boldsymbol{L}}^2|lm\rangle = \hbar^2 l(l+1)|lm\rangle \quad l \in \mathbb{N}$$

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Degeneracy $\quad d_l = 2l + 1$

QM SG
Theoretical Outcome

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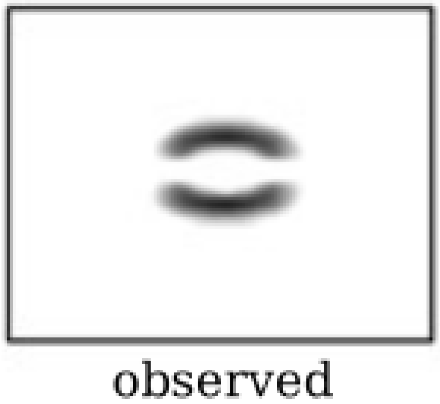

For an electron, $\hat{\boldsymbol{\mu}} = \frac{\hbar q_e}{2m_e} \frac{\hat{\boldsymbol{L}}}{\hbar}$

Theoretical Outcome

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observed

Theoretical Outcome

observed

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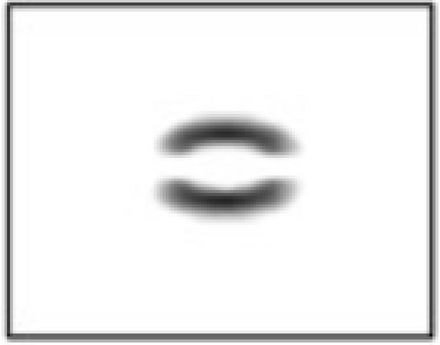

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The energy of the system $|\psi\rangle = |l m_l\rangle$ is

$$\langle \hat{H} \rangle_\psi = -\langle \hat{\mu}_z B_z \rangle_\psi = g_l \mu_B B_z \frac{1}{\hbar} \langle \hat{L}_z \rangle_\psi$$

$\hat{L}_z$ exibits an odd number of discrete eigenvalues $(2l+1)$, so...

Why two deflected spots?

Spin

QM — Intrinsic Angular Momentum

Commutation rules

$$\left[\hat{S}_i, \hat{S}_j\right] = i\hbar\varepsilon_{ijk}\hat{S}_k \qquad \left[\hat{S}_i, \hat{\boldsymbol{S}}^2\right] = 0$$

Basis

The common eigenstates of $\hat{\boldsymbol{S}}^2$ and $\hat{S}_z$ are $|sm_s\rangle$.

Spectrum

$$\hat{\boldsymbol{S}}^2|sm_s\rangle = \hbar^2 s(s+1)|sm_s\rangle$$

$$\hat{S}_z|sm_s\rangle = \hbar m_s|sm_s\rangle \quad -s \leq m_s \leq s$$

Degeneracy

$$d_s = 2s + 1$$

QM — Total Magnetic Momentum

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$$\hat{\boldsymbol{\mu}} = \hat{\boldsymbol{\mu}}_l + \hat{\boldsymbol{\mu}}_s = -\frac{\mu_B}{\hbar}\left(g_l \hat{\boldsymbol{L}} + g_s \hat{\boldsymbol{S}}\right)$$

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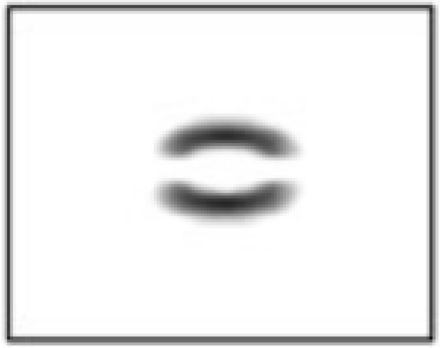
Basis

We can represent the e^- state using the basis

$$\left\{\hat{\boldsymbol{L}}^2, \hat{L}_z, \hat{\boldsymbol{S}}^2, \hat{S}_z\right\} \quad \Rightarrow \quad |\psi\rangle \equiv |l m_l\rangle \otimes |s m_s\rangle$$

Experimental Outcome



observed

Energy Spectrum

$$E_{m_l m_s} = \frac{\mu_B B_z}{\hbar}\left(\langle \hat{L}_z \rangle_\psi + 2\langle \hat{S}_z \rangle_\psi\right)$$
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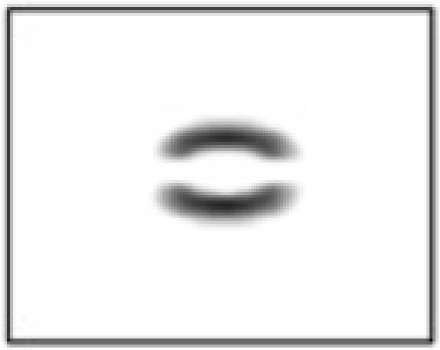

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Degeneracy $d = (2l+1)(2s+1)$

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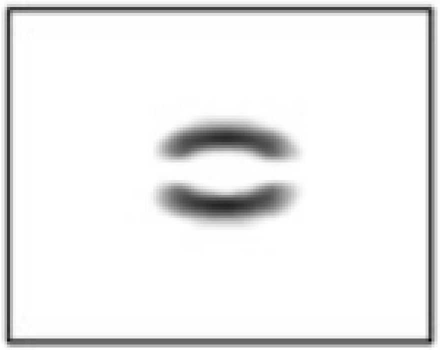

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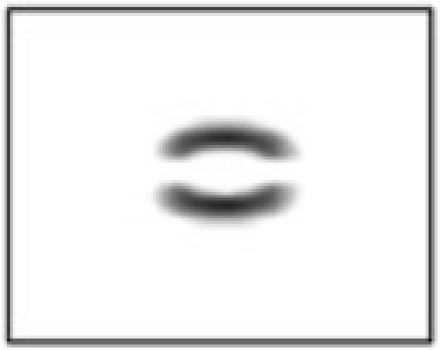

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Also, to make the deflection magnitude match the experiment $g_s = 2$

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Consider a state $|\psi\rangle = |lm_l\rangle \otimes |sm_s\rangle$

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$$\left[\hat{J}_i, \hat{J}_j\right] = i\hbar\varepsilon_{ijk}\hat{J}_k \qquad \left[\hat{J}_i, \hat{\boldsymbol{J}}^2\right] = 0$$

Basis

The common eigenstates of $\hat{\boldsymbol{J}}^2$, $\hat{J}_z$ are $|jm_j\rangle$.

Total Angular Momentum

Consider a state $|\psi\rangle = |lm_l\rangle \otimes |sm_s\rangle$

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Coupling of Angular Momenta

Consider the magnetic field experienced by an e^- orbiting around a nucleus with $Z\ p^+$

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Another option is the **coupled** basis

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since

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Energy Spectra in Magnetic Field

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{s-o}} + \frac{\mu_B B_z}{\hbar}\left(\hat{L}_z + 2\hat{S}_z\right)$$

Energy Spectra in Magnetic Field

Paschen-Back Effect $\quad \mu_B B \gg \xi$

$$\xi_{m_l m_s} = \frac{\mu_B B_z}{\hbar}\left(\langle \hat{L}_z \rangle_\psi + 2\langle \hat{S}_z \rangle_\psi\right)$$

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- Spin-orbit coupling negligible
- L and S precess independently around B
- Good quantum numbers: m_l, m_s
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Landé g-factor

$$g_j = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}$$

Boson & Fermions

Hello world

Conclusion

Hello world